

A geometric experiment

Repeat independent trials with constant success probability p until the first success.

How does the stopping rule differ from a binomial experiment?

Our counting convention

X counts trials through the first success, including the successful trial.

What is X for failure, failure, success?

Possible values

X counts trials through the first success.

Can X equal zero?

Identify p

Each independent roll of a fair die continues until the first 6.

What is the success probability p ?

Geometric notation

X follows a geometric distribution when it counts trials through the first success.

$$X \sim G(p)$$

What experiment could $X \sim G(0.2)$ describe?

First success on trial four

Each independent trial succeeds with probability 0.2.

What sequence is required for the first success on trial four?

The geometric probability

For a first success on trial x , the preceding $x - 1$ trials must fail.

$$P(X = x) = (1 - p)^{x-1}p$$

For $p = 0.2$, what is $P(X = 4)$?

Success immediately

Geometric X counts trials through first success; $p = 0.2$.

What is $P(X = 1)$?

More than three trials

Geometric X counts trials through first success; $p = 0.2$.

What must happen in the first three trials for $X > 3$?

A tail probability

Geometric X counts trials through first success; $p = 0.2$.

What is $P(X > 3)$?

Success by trial three

Geometric X counts trials through first success; $p = 0.2$.

How can a complement calculate $P(X \leq 3)$?

At least three trials

Geometric X counts trials through first success; $p = 0.2$.

How many initial failures are required for $X \geq 3$?

Expected trials

A geometric trial count has mean $1/p$.

$$\mu = \frac{1}{p}$$

For $p = 0.2$, how many trials are expected?

Standard deviation

A geometric trial count has standard deviation $\sqrt{(1 - p)/p}$.

$$\sigma = \frac{\sqrt{1 - p}}{p}$$

For $p = 0.2$, what is σ ?

Expectation is not a deadline

A geometric model has expected trial count 5.

Must the first success occur by trial five?

Memoryless waiting

Geometric X counts trials through success; $0 < p < 1$ and integer $m, k \geq 0$.

$$P(X > m + k \mid X > m) = P(X > k)$$

After four failures, does the distribution of additional trials change?

Additional waiting

Independent trials have $p = 0.2$. Four trials have already failed.

What is the probability that the next two trials also fail?

Success is not “due”

Independent trials have $p = 0.2$. Ten trials have already failed.

What is the chance the next trial succeeds?

Count failures instead

Let Y count failures before the first success, while X counts trials through that success.

How are X and Y related?

Failures-before-success probabilities

Y counts failures before the first success, so $Y = 0, 1, 2, \dots$. Each independent trial succeeds with probability p .

$$P(Y = y) = (1 - p)^y p$$

For $p = 0.2$, what is $P(Y = 3)$?

Expected failures

For failures before success, $E(Y) = (1 - p)/p$.
Trials through success have $E(X) = 1/p$.

For $p = 0.2$, what are the two expected counts?

Spread under a change of convention

Trial count X and failure count Y satisfy $Y = X - 1$. Both have variance $(1 - p)/p^2$.

Why does subtracting one change the mean but leave the standard deviation unchanged?

The common ratio

Geometric probabilities form a geometric sequence: each probability is $1 - p$ times the preceding one. This factor is the common ratio.

For $p = 0.2$ and $P(X = 3) = 0.128$, what is $P(X = 4)$?

Graphing geometric probabilities

Let $X \sim G(0.25)$ count trials through first success.

How would the first five probability bars show the geometric progression?

When the model changes

A user learns from each attempt, increasing the chance of success.

Which geometric assumption is no longer appropriate?